

1.Stack

```
import java.util.*;

class main{

    public static void main(String[]args){

        int n=5;

        int pushed[]={1,2,3,4,5};

        int popped[]={5,4,3,2,1};

        int index=0;

        Stack <Integer> s=new Stack<>();

        for(int i=0;i<n;i++){

            s.push(pushed[i]);

            while(!s.isEmpty() && s.peek()==popped[index]){

                s.pop();

                index++;

            }

        }

        if(s.isEmpty()){

            System.out.print("true");

        }

        else{

            System.out.print("flase");

        }

    }

}
```